

**TABLE 7**

Percentage points of the chi-square distribution

df	$\alpha =$	.999	.995	.99	.975	.95	.90
1		.000002	.000039	.000157	.000982	.003932	.01579
2		.002001	.01003	.02010	.05064	.1026	.2107
3		.02430	.07172	.1148	.2158	.3518	.5844
4		.09080	.2070	.2971	.4844	.7107	1.064
5		.2102	.4117	.5543	.8312	1.145	1.610
6		.3811	.6757	.8721	1.237	1.635	2.204
7		.5985	.9893	1.239	1.690	2.167	2.833
8		.8571	1.344	1.646	2.180	2.733	3.490
9		1.152	1.735	2.088	2.700	3.325	4.168
10		1.479	2.156	2.558	3.247	3.940	4.865
11		1.834	2.603	3.053	3.816	4.575	5.578
12		2.214	3.074	3.571	4.404	5.226	6.304
13		2.617	3.565	4.107	5.009	5.892	7.042
14		3.041	4.075	4.660	5.629	6.571	7.790
15		3.483	4.601	5.229	6.262	7.261	8.547
16		3.942	5.142	5.812	6.908	7.962	9.312
17		4.416	5.697	6.408	7.564	8.672	10.09
18		4.905	6.265	7.015	8.231	9.390	10.86
19		5.407	6.844	7.633	8.907	10.12	11.65
20		5.921	7.434	8.260	9.591	10.85	12.44
21		6.447	8.034	8.897	10.28	11.59	13.24
22		6.983	8.643	9.542	10.98	12.34	14.04
23		7.529	9.260	10.20	11.69	13.09	14.85
24		8.085	9.886	10.86	12.40	13.85	15.66
25		8.649	10.52	11.52	13.12	14.61	16.47
26		9.222	11.16	12.20	13.84	15.38	17.29
27		9.803	11.81	12.88	14.57	16.15	18.11
28		10.39	12.46	13.56	15.31	16.93	18.94
29		10.99	13.12	14.26	16.06	17.71	19.77
30		11.59	13.79	14.95	16.79	18.49	20.60
40		17.92	20.71	22.16	24.43	26.51	29.05
50		24.67	27.99	29.71	32.36	34.76	37.69
60		31.74	35.53	37.48	40.48	43.19	46.46
70		39.04	43.28	45.44	48.76	51.74	55.33
80		46.52	51.17	53.54	57.15	60.39	64.28
90		54.16	59.20	61.75	65.65	69.13	73.29
100		61.92	67.33	70.06	74.22	77.93	82.36
120		77.76	83.85	86.92	91.57	95.70	100.62
240		177.95	187.32	191.99	198.98	205.14	212.39

TABLE 7
(continued)

$\alpha = .10$	.05	.025	.01	.005	.001	df
2.706	3.841	5.024	6.635	7.879	10.83	1
4.605	5.991	7.378	9.210	10.60	13.82	2
6.251	7.815	9.348	11.34	12.84	16.27	3
7.779	9.488	11.14	13.28	14.86	18.47	4
9.236	11.07	12.83	15.09	16.75	20.52	5
10.64	12.59	14.45	16.81	18.55	22.46	6
12.02	14.07	16.01	18.48	20.28	24.32	7
13.36	15.51	17.53	20.09	21.95	26.12	8
14.68	16.92	19.02	21.67	23.59	27.88	9
15.99	18.31	20.48	23.21	25.19	29.59	10
17.28	19.68	21.92	24.72	26.76	31.27	11
18.55	21.03	23.34	26.22	28.30	32.91	12
19.81	22.36	24.74	27.69	29.82	34.53	13
21.06	23.68	26.12	29.14	31.32	36.12	14
22.31	25.00	27.49	30.58	32.80	37.70	15
23.54	26.30	28.85	32.00	34.27	39.25	16
24.77	27.59	30.19	33.41	35.72	40.79	17
25.99	28.87	31.53	34.81	37.16	42.31	18
27.20	30.14	32.85	36.19	38.58	43.82	19
28.41	31.41	34.17	37.57	40.00	45.31	20
29.62	32.67	35.48	38.93	41.40	46.80	21
30.81	33.92	36.78	40.29	42.80	48.27	22
32.01	35.17	38.08	41.64	44.18	49.73	23
33.20	36.42	39.36	42.98	45.56	51.18	24
34.38	37.65	40.65	44.31	46.93	52.62	25
35.56	38.89	41.92	45.64	48.29	54.05	26
36.74	40.11	43.19	46.96	49.65	55.48	27
37.92	41.34	44.46	48.28	50.99	56.89	28
39.09	42.56	45.72	49.59	52.34	58.30	29
40.26	43.77	46.98	50.89	53.67	59.70	30
51.81	55.76	59.34	63.69	66.77	73.40	40
63.17	67.50	71.42	76.15	79.49	86.66	50
74.40	79.08	83.30	88.38	91.95	99.61	60
85.53	90.53	95.02	100.43	104.21	112.32	70
96.58	101.88	106.63	112.33	116.32	124.84	80
107.57	113.15	118.14	124.12	128.30	137.21	90
118.50	124.34	129.56	135.81	140.17	149.45	100
140.23	146.57	152.21	158.95	163.65	173.62	120
268.47	277.14	284.80	293.89	300.18	313.44	240

Source: Computed by P. J. Hildebrand.